Protocol - Expression Induction in Transformed Bacteria

TEMPLATE · IGEM 2026 CONTRIBUTION

What you're going to get / What do you expect

This protocol grows transformed E. coli BL21(DE3) to the exponential phase and induces expression of the target enzyme with IPTG. One overnight pre-inoculum is enough to start a 50 mL induced culture, which is then harvested following Protocol 4.

Materials list

- Transformed *E. coli* BL21(DE3) colonies (from the transformation plates)
- LB components: yeast extract, tryptone, NaCl, distilled water
- **Kanamycin** (50 mg/mL stock; **50 µg/mL final**)
- **IPTG, 1 M stock** (238 mg/mL)
- Autoclave
- Erlenmeyer flasks
- Shaking incubator (**37 °C** and **30 °C**)
- Spectrophotometer and 1 mL cuvettes (OD_{600})
- Micropipettes and sterile tips

Steps

1. Prepare LB medium (**yeast extract 5 g/L, tryptone 10 g/L, NaCl 10 g/L**, made up to 1 L with distilled water).
2. Autoclave the LB medium and add kanamycin (**50 µg/mL final**) once it has cooled.
3. Prepare the pre-inoculum. Pick a colony from the transformation plate and inoculate into **5 mL of LB with kanamycin**. Incubate **overnight at 37 °C** with shaking.
4. Inoculate the main culture. Transfer **1 mL of the pre-inoculum** into a larger flask with LB and kanamycin (for example 1 mL into **50 mL**) and grow at **37 °C, 180 rpm**.
5. Follow bacterial growth so that induction falls in the exponential phase, when transcriptional and translational activity is highest.
6. Measure the optical density at 600 nm over time, against an LB blank, until it reaches **OD_{600} 0.5–0.7**.
7. Induce expression by adding IPTG to a final concentration of about **0.4 mM**.
8. Lower the temperature to **30 °C** and incubate the induced culture **overnight** with shaking at **140 rpm**.

Troubleshooting

- **Low or no expression:** Confirm that induction took place within **OD_{600} 0.5–0.7** and that kanamycin was present throughout, so that the plasmid is retained.

- **Insoluble protein:** Lowering the temperature after induction favours soluble expression. If the enzyme still appears in the insoluble fraction, test a lower temperature or a lower IPTG concentration.
- **Antibiotic degradation:** Add kanamycin only after the autoclaved medium has cooled.
- **IPTG volume:** For a 50 mL culture, **20 µL of 1 M IPTG** gives 0.4 mM final.